



Cutting-edge computational approaches in enzyme design and activity enhancement

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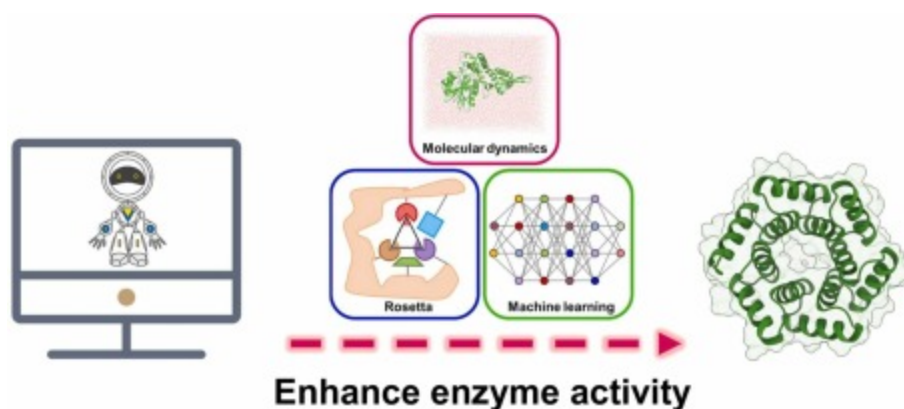
Highlights

- MD simulations boost enzyme activity by revealing dynamic conformational changes.
- Rosetta-based energy ranking enables efficient enzyme activity prediction.
- Machine learning significantly reduces the screening workload in enzyme evolution.
- Computation-driven enzyme design marks a new phase in enzyme engineering.

Abstract

Enzyme activity is crucial in biocatalysis, making methods to enhance enzyme performance a major focus of research. Computational design provides an efficient approach to boosting enzyme activity, thereby expanding its applications across various fields. This review highlights three main computational methods: molecular dynamics simulations, Rosetta, and machine learning, and explores recent advances in their use for rapidly enhancing enzyme activity in enzyme engineering. These techniques provide a novel perspective on enzyme activity optimization, significantly reducing the complexity of traditional screening processes. By integrating these advanced computational approaches, high-activity enzymes can be designed more rapidly, accelerating progress in protein engineering and synthetic biology.

Graphical Abstract



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Introduction

Enzymes, as biological catalysts, are crucial for accelerating biochemical reactions without being consumed or altered in the process[1], [2]. Enhancing enzyme activity is essential across multiple sectors, including industrial production, biomedicine, and environmental protection. For instance, improving production efficiency, reducing costs, and minimizing pollutant emissions can expand the applications of enzymes in the pharmaceutical, food, and energy sectors[3], [4].

In the field of enzyme engineering, directed evolution is a prominent method for enhancing enzyme activity. Depending on the method of mutant library construction, directed evolution can be categorized into non-rational, semi-rational, and rational design

approaches[5], [6]. Non-rational design primarily includes techniques such as saturation mutagenesis, error-prone PCR, and DNA recombination. Semi-rational design involves methods like combinatorial active-site saturation testing (CAST) and iterative saturation mutagenesis (ISM). In contrast, rational design leverages computational technologies to significantly reduce the screening scale, allowing precise customization of target enzymes within a limited sequence space. Over the years, semi-rational and non-rational designs have achieved remarkable success in enhancing enzyme activity. However, computer-aided rational design has rapidly evolved and has become a crucial strategy for improving enzyme performance. Unlike traditional non-rational and semi-rational approaches, rational design relies on computational simulation techniques, particularly MD simulations, Rosetta molecular modeling software, and machine learning (ML) algorithms, which are extensively used to accurately predict and tailor enzyme functions. By employing these computational strategies, researchers can rapidly identify target mutation sites within a confined sequence space and design enzyme variants with superior performance, thereby reducing the scale and cost of experimental screening. These advancements highlight the transformative role of computational approaches in enzyme engineering, paving the way for more efficient and precise enzyme design.

MD simulations are essential tools in rational enzyme design, as they provide detailed insights into enzyme-substrate interactions and conformational changes at the atomic level[7]. MD simulations allow for the exploration of enzyme structures, revealing the dynamic pre-organization of active sites and identifying key residues that contribute to enzyme activity[8], [9], [10]. Strategies based on MD simulations for enhancing enzyme activity primarily include tunnel engineering and conformation-driven regulation of distal sites. Tunnel engineering focuses on the design and optimization of substrate entry or product release tunnels[11], [12]. MD simulations can identify the positions and roles of key residues within the tunnel, predicting how mutations can alter the size, shape, or surface charge of the tunnel to optimize substrate binding and transport efficiency. Conformation-driven regulation of distal sites involves altering residues far from the active site to indirectly influence the conformation and function of the catalytic center[13]. MD simulations reveal how these distal sites can affect catalytic activity through conformational coupling or dynamic networks. Furthermore, enhanced sampling techniques in MD simulations can accelerate the capture of rare events, providing broader conformational sampling and improving the accuracy of enzyme design[14], [15].

Rosetta is a highly flexible computational tool that integrates various functions such as de novo protein design, enzyme active site design, molecular docking, and macromolecular structure prediction. In the context of enhancing enzyme activity, the most commonly used

module is Rosetta Design, which optimizes the tertiary structure and active site models of enzymes and rapidly assesses the impact of different mutations on enzyme activity[16], [17]. A typical approach to enzyme design using Rosetta involves grafting transition state structures onto natural protein scaffolds. Additionally, Rosetta has given rise to numerous enzyme design strategies and software tools, with FuncLib and pSUFER being notably effective in enhancing enzyme activity[18], [19]. This enhancement is primarily attributed to improvements in protein folding, which subsequently lead to increased catalytic efficiency.

In recent years, ML has emerged as a powerful tool in enzyme design and activity enhancement[20], [21]. ML can predict protein structures, improve enzyme functional properties, and guide the exploration of directed evolution libraries. Unlike model-driven rational design or traditional directed evolution, ML is data-driven, capable of analyzing large-scale experimental data to extract key features and predict unseen enzyme variants. This significantly enhances prediction efficiency and uncovers design spaces that traditional methods often overlook. Currently, the integration of ML with directed evolution is widely used, greatly reducing the extensive efforts typically associated with traditional evolutionary approaches[22]. ML is gradually becoming an indispensable tool in enzyme engineering, providing novel insights and technological support for future rational enzyme design.

This paper provides an overview of the general strategies for computer-aided enzyme customization developed in recent years, focusing primarily on the cutting-edge computational approaches of MD simulations, Rosetta, and ML in enhancing enzyme activity. The aim is to achieve rapid improvements in enzyme performance with minimal design and screening efforts through computer-driven engineering. The applications of these methods in various fields, including industry, pharmaceuticals, and environmental science, demonstrate their powerful design capabilities and vast potential for future use.

Section snippets

Molecular dynamics simulation

The fine-tuning of enzyme conformational ensembles plays a significant role in molecular innovation and the evolution of new catalytic functions, highlighting the critical importance of conformational dynamics in enzyme evolution. MD simulations, often referred to as a “virtual microscope,” are widely used to study enzyme conformational dynamics. ...

Methods based on the Rosetta platform

Rosetta plays a critical role in enzyme design by simulating and optimizing enzyme structures to enhance catalytic efficiency, stability, and substrate specificity, making it a core tool in the field. The classic “Inside-out” design strategy, which is based on transition state enzyme design, has been widely applied to improve enzyme activity. Additionally, various enzyme design algorithms and tools have been developed based on Rosetta, including PROSS for optimizing protein stability, Rosetta ...

Applications of machine learning in enhancing enzyme activity

ML is an emerging method in the field of enzyme design, with research on enzyme-related ML experiencing exponential growth over the past decade (Fig. 5A). Compared to traditional methods, ML is a data-driven strategy that can parse existing data through various descriptors and use multiple model algorithms to predict properties of previously unseen but similar inputs (Fig. 5B)[20]. Depending on the representation of proteins, training data, and algorithms used, ML models in protein design can ...

Conclusion

In practical applications, a single strategy often fails to significantly enhance enzyme activity, whereas the combination of multiple strategies through computational simulation can maximize the evolution of enzymes and reduce screening risks. For instance, in the early stages of enzyme evolution, if crystal structures are unavailable, tools such as AlphaFold2[86], AlphaFold3[90] and RosettaFold[87] can be utilized to predict the three-dimensional structures of enzymes. The enzyme's binding ...

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CRedit authorship contribution statement

Pu Zheng: Writing – review & editing, Project administration, Funding acquisition.

Pengcheng Chen: Supervision, Methodology, Investigation. **Dan Wu:** Supervision,

Methodology, Investigation. **Ruobin Sun:** Writing – original draft, Methodology, Investigation, Conceptualization. ...

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. ...

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